Raspberry Pi Pico

+3.3V
220
R1
3V1
green
GND

R10
220k
GND

+BATT
GND
+3.3V
J1
5V

NT5 NetTie_2
NT6 NetTie_2
NT7 NetTie_2
NT8 NetTie_2

P1
rpi pico 1
2648-SC0915CT-ND
tht

5V
D1
D_Schottky
+3.3V

VBUS
VSYS
3V3_OUT
ADC_VREF
3V3_EN
RUN
GP0
GP1
GP2
GP3
GP4
GP5
GP6
GP7
GP8
GP9
GP10
GP11
GP12
GP13
GP14
GP15
GP16
GP17
GP18
GP19
GP20
GP21
GP22
GP26
GP27
GP28
AGND
GND
GND__1
GND__2
GND__3
GND__4
GND__5
GND__6
GND

bt1
bt2
bt3
bt4
sda
scl
right_2
right_1
right_en
left_2
left_en
left_1

R2
220
D13
blue
GND

+3.3V
J3
joystick
GND

+3.3V
J4
speed knob
GND

left motor

U1
DRV8245HQPWPRQ

LVL6, in independent mode that means faults are retries

SR
DIAG
PH/IN2
EN/IN1
DRVOFF
VM
OUT1
OUT2
GND
EPAD
ITRIP
MODE
NFAULT
IPROPI
NSLEEP

left_2
left_1
left_ipropi
left_en

+BATT
GND
C1
0.1uF

8.2k +-10% LVL2 INDEPENDENT
8.2k
R7
GND

HSL1
Heatsink

J100
0197054001
J101
0197054001
optionally replace J7 to be weelye compatible

ML+
ML-
J7
1017491

U2
DRV8245HQPWPRQ

right_1
right_2
right_ipropi
right_en

C2
0.1uF

8.2k +-10% LVL2 INDEPENDENT
8.2k
R6

HSR1
Heatsink

MR-
MR+
J8
1017491

optionally replace J8 to make the PCB weelye compatible
wellye style
J104
wellye RX25 style
J105

ADC (for measuring battery voltage and motor current)

+3.3V
NT1
NetTie_3
U4
TLA2024IRUGR
+3.3V

+BATT
R13
100k
R14
8.2k
GND

TP0
TestPoint
ADDR
AIN0
AIN1
AIN2
AIN3
NC
VDD
SCL
SDA
GND

right_ipropi
left_ipropi
scl
sda

C4
0.1uF
GND

pullup resistors

TP1
TestPoint
left_ipropi
TP2
TestPoint
right_ipropi
R11
604
R12
604
GND
GND

+3.3V
R15 3.3k
R16 3.3k
sda
scl

qwiic I2C plug

GND
+3.3V
sda
scl
J2
Conn_01x04
MP

weelye compatible header

J11
+BATT
switch
bat-
D2
D_Schottky
+BATT
Conn_01x10_Pin

3.5mm audio jacks for if you want to connect buttons as an alternative to the joystick

B1
ASJ-200-B-HT
B2
ASJ-200-B-HT
B3
ASJ-200-B-HT
B4
ASJ-200-B-HT
bt1
bt2
bt3
bt4
GND

battery power circuit

J102
0197054001
optionally replace J5 to be weelye compatible
J103
0197054001
J5
TB005-762-02BE
bat-

R3
3.3k
SW1
screw terminal
C5
33uf
GND
+BATT
18V zener
D3
13.7k
R4
switch
J6

PSMN1R2-25YLDX
NMOS
Q1
reverse polarity protection
PSMN1R2-25YLDX
Q2
on/off switch
GND

3mm diameter bolt holes
3.35in x 2.2in
85.09mm x 55.88mm
H1
M3
H2
M3
H3
M3
H4
M3

alternative voltage regulator

Instead of the voltage regulator circuit, you could get a single THT module that outputs 5V.
Make sure the pinout matches.
Make sure the min and max input voltage are compatible with your battery.

U5
D78B05T, RBT05W24S05, pololu5593
+BATT
IN
GND
OUT
5V
GND

voltage regulator circuit https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=F7F7D983B50C08E2

+BATT
R5
100k
U3
TPS54202HDDCR
EN
FB
VIN
BOOT
SW
GND
C6
0.1uF
L1
12uH
5V
C7
56pF
R8
100k
C8
47uF
C3
10uF
R9
13.7k
GND

https://github.com/gobabygocarswithjoysticks gobabygocarswithjoysticks@gmail.com

Sheet: /
File: gbg-pcb.kicad_sch

Title:

Size: A3
Date:
Rev:
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